

A Short Proof of the Quasi-Riemann Hypothesis

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Abstract

We prove that the Riemann zeta function, all Dirichlet L -functions, and all finite-order Hecke L -functions over $\mathbb{Q}(\sqrt{-3})$ are zero-free in $\operatorname{Re} s > 47/48$. Our proof combines cubic theta reflection, a quadratic large sieve, and Hecke zero-density estimates with Mellin continuation.

1 Introduction

The Quasi-Riemann Hypothesis asserts that every nontrivial zero ρ of $\zeta(s)$ satisfies $\operatorname{Re} \rho \leq 1 - c$ for some fixed $c > 0$. We give a short proof with $c = 1/48$.

Theorem 1. *The Riemann zeta function has no zeros in $\operatorname{Re} s > 47/48$. More generally, every finite-order Hecke L -function over $F = \mathbb{Q}(\sqrt{-3})$, and every Dirichlet L -function, is zero-free in this half-plane. Principal characters have their usual pole at $s = 1$.*

1.1 Proof overview

The Mellin mechanism. Let $f : (0, \infty) \rightarrow \mathbb{C}$ be continuous. If $f(Z) = O(Z^{r-2/3})$ as $Z \rightarrow \infty$ and $f(Z) = O_M(Z^{M-2/3})$ as $Z \rightarrow 0$ for every $M > 2$, then

$$\int_0^\infty f(Z) Z^{-s+2/3} \frac{dZ}{Z}$$

is holomorphic for $\operatorname{Re} s > r$. At infinity the integrand is bounded by a constant times $Z^{r-\operatorname{Re} s} dZ/Z$; at zero we choose $M > \operatorname{Re} s$. Thus a growth bound for an inverse Mellin transform can continue its transform beyond the original domain.

The three proof obligations. Suppose, for a contradiction, that $\beta_* > b$, and fix a primitive target η . We construct a normalized probe J_η and a signal

$$f_\eta(Z) = \frac{1}{2\pi i} \int_{(2)} Z^{s-2/3} e^{(s-5/6)^2} \frac{H_\eta(s)}{L_F^S(s, \eta)} ds,$$

where S is a finite set of excluded primes. We prove the following; the estimates involving Z hold for $Z \geq 2$:

1. **A small probe:** $J_\eta(Z) \ll_{\eta, \varepsilon} Z^{1/4+\varepsilon}$ (Section 3).
2. **An accurate comparison:** $J_\eta(Z) - f_\eta(Z) \ll_\eta Z^{\beta_*-2/3-\delta_0}$ (Sections 4, 6, and 7).
3. **A nonvanishing numerator:** H_η is holomorphic and $|H_\eta(s)| \geq 1/2$ on $\operatorname{Re} s > 11/12$ (Section 5).

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How these exclude zeros. Take $\varepsilon = 1/32$. Since $\beta_* - 2/3 - \delta_0 > 9/32$, the first two estimates give $f_\eta(Z) \ll_\eta Z^{\beta_* - 2/3 - \delta_0}$. Its defining contour also gives the required decay at zero. The Mellin mechanism therefore continues its transform to $\operatorname{Re} s > \beta_* - \delta_0$. Mellin inversion identifies this transform with $e^{(s-5/6)^2} H_\eta(s)/L_F^S(s, \eta)$ on $\operatorname{Re} s > 1$. The third estimate permits division by the numerator, continuing $1/L_F^S(s, \eta)$ holomorphically to $\operatorname{Re} s > \beta_* - \delta_0$.

The same $\delta_0 > 0$ works for every target. By the definition of the supremum, some target has a zero in that half-plane, where its reciprocal must have a pole. This is the contradiction. Section 8 supplies the full continuation argument.

1.2 Notation

Write $\omega = e^{2\pi i/3}$, $\mathcal{O} = \mathbb{Z}[\omega]$, and $\lambda = 1 + 2\omega$. Write $q_a = N(a) = |a|^2$ for elements, and use the same norm notation for ideals. The ring is norm-Euclidean; every ideal prime to three has a unique generator congruent to one modulo $3\mathcal{O}$, called primary. The unit group is $\mu_6 = \{\pm 1, \pm\omega, \pm\omega^2\}$. Lattice counting gives $O(X)$ ideals or elements of norm at most X . Set $\alpha(a) = a/|a|$ and $e(z) = \exp(2\pi i \operatorname{Tr}_{F/\mathbb{Q}}(z/\lambda))$. For planar Fourier transforms, extend the trace by $\operatorname{Tr}(w) = w + \bar{w}$ for $w \in \mathbb{C}$. The measure $d\mu(z) = (2/\sqrt{3}) dx dy$ makes $\mathcal{O}$ self-dual for the pairing $e(zy)$; our transform is $\mathcal{F}_{\mathbb{C}} f(y) = \int_{\mathbb{C}} f(z) e(-zy) d\mu(z)$.

For a good prime $\mathfrak{p}$, define $\chi_{\mathfrak{p}}(a) = (a/\mathfrak{p})_6$ by reduction of $a^{(q_{\mathfrak{p}}-1)/6}$ into μ_6 . Extend multiplicatively in a primary denominator. Every character power, including exponent zero or a negative exponent, is zero on nonunits. We distinguish the two orientations of the symbol: $\chi_{\mathfrak{a}}(v)$ is the residue character in the element v , whereas $\psi_u(\mathfrak{n}) = \chi_{\mathfrak{n}}(u)$ is the Kummer Hecke character in the ideal $\mathfrak{n}$. In expressions such as $\chi_R(n)$, the argument is the chosen primary generator of the ideal n . The excluded set S contains the primes above six, the target conductor primes, and a fixed small-prime cutoff specified below. A good prime lies outside S . Unqualified sums over a, b, c, n, s are ideal sums represented by their unique primary generators and avoid S , unless expressly stated otherwise. The variables m and u range over elements, including their unit factors. The superscript sf means squarefree, and $(a, S) = 1$ means that no prime in S divides a . Superscript S on an L -function means deletion of the Euler factors in S . We use the Mellin transform $\widehat{W}(t) = \int_0^\infty W(r) r^{t-1} dr$; an integral over (σ) runs upward on $\operatorname{Re} t = \sigma$. An annular smooth weight belongs to $C_c^\infty((0, \infty))$. Implied constants may depend on the fixed target, S , the weights, and displayed small parameters; uniformity in moving rows and scales will be stated where needed.

2 One probe and its two exact representations

The probe is chosen to admit two useful descriptions. Its cubic Gauss coefficients match the theta reflection formula, giving the low estimate. Poisson summation gives an Euler product whose principal row contains the reciprocal of the target L -function, giving the signal. Smooth annular weights and a Mellin Gaussian make both operations convergent and control their tails. The finite phase correction below makes the arithmetic in these descriptions agree.

Put

$$g_\psi(c, k) = \sum_{d \bmod c} \psi(d) e(kd/c), \quad \gamma_j(c) = q_c^{-1/2} g_{\chi_c^j}(c, 1) \quad (c \text{ squarefree}).$$

At modulus one these functions equal one.

A finite phase correction. For odd $c = a + b\omega$, define

$$\Gamma_q(c) = \frac{1 + i^{-b} + i^a + i^{b-a}}{2}, \quad G(c) = \overline{\chi_c(4)}\Gamma_q(c), \quad R(v, w) = \frac{\Gamma_q(vw)}{\Gamma_q(v)\Gamma_q(w)}. \quad (1)$$

Here $G(c)$ is used only for primary c prime to six. The functions G and R factor through a fixed finite ray group $\mathcal{T}$ supported at 2, 3; a sufficiently divisible modulus, for example 36, includes the primary and supplementary conventions. Appendix A proves the phase identities that make the following two representations agree.

The physical probe. Enlarge S to include the target's conductor and every prime of norm at most a fixed P_0 . Let b_* generate its radical. Choose a character ξ modulo b_* nonprincipal at every prime, of order dividing six: use a quadratic character at odd residue characteristic and a cubic character at the prime of norm four. Define

$$\tau = q_{b_*}^{-1/2} g_\xi(b_*, 1), \quad \Xi(A) = \xi(A)\chi_A(b_*), \quad |\tau| = 1.$$

Here Ξ is a fixed finite ray character supported on S . For $A = cn^3$ set

$$T_{m,s,\eta}(t) = \sum_{c \text{ sf}, n} \gamma_2(c) \overline{\alpha(A)} \overline{G(A)} \eta(A) \Xi(A)^{-1} R(A, s) \chi_A(m) q_c^{-1/2-t} q_n^{-1-3t}.$$

Choose nonnegative nonzero smooth annular W_0, W_1 and put $\Phi(t) = e^{t^2}$. The probe is the following explicitly convergent expression:

$$I_\eta(X, Y, Z) = \frac{1}{Y} \sum_s \frac{W_1(q_s/Y) \chi_s(b_*)}{\tau \xi(s) \sqrt{q_{b_*} q_s X}} \sum_{m \in \mathcal{O}} \xi(m) q_s^{-1/2} g_{\chi_s}(s, -m) W_0\left(\frac{q_m}{q_{b_*} q_s X}\right) \times \frac{1}{2\pi i} \int_{(4)} Z^t \Phi(t) T_{m,s,\eta}(t) dt. \quad (2)$$

All inverse character values are evaluated only on elements prime to S .

Exact low separation. For a class $\sigma \in \mathcal{T}$, choose a representative s_σ . Use the finite Fourier expansion and character

$$\overline{G(A)} = \sum_{\theta \in \widehat{\mathcal{T}}} a_\theta \theta(A), \quad \nu_\sigma(A) = \eta(A) \Xi(A)^{-1} R(A, s_\sigma).$$

Put

$$B_{m,\sigma}(Z) = \sum_\theta a_\theta C_{V_G}(Z; m, \nu_\sigma \theta), \quad V_G(x) = \frac{e^{-(\log x)^2/4}}{2\sqrt{\pi}}.$$

Here C_V is the completed row defined in (7). The characters $\nu_\sigma \theta$ form a fixed finite family supported on S , so the completed-row estimate applies with every original zero mask. For $Q_0 = q_{b_*} XY$ define

$$A_{m,\sigma,v}(Y) = \frac{1}{Y} \sum_{s \in \sigma} \frac{W_1(q_s/Y) \chi_s(b_*)}{\tau \xi(s)} (q_s/Y)^{-1/2+iv} q_s^{-1/2} g_{\chi_s}(s, -m).$$

Choose a compactly supported Ω equal to one on the original range of q_m/Q_0 . Mellin inversion gives the exact identity

$$I_\eta = \frac{Q_0^{-1/2}}{2\pi} \sum_\sigma \int_{\mathbb{R}} \widehat{W}_0(iv) \sum_{m \neq 0} \Omega(q_m/Q_0) \xi(m) (q_m/Q_0)^{-iv} A_{m,\sigma,v}(Y) B_{m,\sigma}(Z) dv. \quad (3)$$

Indeed,

$$\frac{q_m}{q_{b*} q_s X} = \frac{q_m/Q_0}{q_s/Y}, \quad (q_{b*} q_s X)^{-1/2} = Q_0^{-1/2} (q_s/Y)^{-1/2}.$$

Mellin inversion gives (3), and annular support removes $m = 0$.

The additive norm. The Gauss expansion writes $A_{m,\sigma,v} = \sum_{s,d} a_{s,d} e(-md/s)$, where d is a unit modulo s , $q_s \asymp Y$, and $|a_{s,d}| \ll Y^{-3/2}$. These fractions are distinct: equality modulo $\mathcal{O}$ implies $s \mid s'$ and $s' \mid s$, hence equality of the primary denominators and residue classes. Distinct fractions are separated by $\delta \gg Y^{-1}$, since their difference has a nonzero integral numerator and denominator of absolute value $O(Y)$. There are $O(Y^2)$ pairs, so $\sum |a_{s,d}|^2 \ll Y^{-1}$, uniformly in v . To bound their row norm, majorize the ball $q_m \leq CQ$ by a fixed multiple of $e^{-cq_m/Q}$. Poisson summation gives a Gram kernel bounded in absolute value by $C'Q \sum_{n \in \mathcal{O}} e^{-c'Q|n-z_j+z_k|^2}$. The δ -separated lifts have at most $O(1+r^2/\delta^2)$ points in a radius- r disk; summing Gaussian shells bounds every matrix row sum by $O(Q + \delta^{-2})$. The elementary Schur bound therefore gives

$$\sum_{q_m \leq CQ} |A_{m,\sigma,v}(Y)|^2 \ll \frac{Q + Y^2}{Y}. \quad (4)$$

Exact Poisson separation. Let $\widetilde{W}_0(|y|^2)$ be the planar Fourier transform of $W_0(|y|^2)$ and set

$$M(z) = \int_0^\infty \widetilde{W}_0(r) r^{z-1} dr, \quad \widehat{W}_1(w) = \int_0^\infty W_1(r) r^{w-1} dr.$$

The same probe has the following representation, where u ranges over nonzero sixth-power-free elements prime to S , including all six units. With all three integrals initially absolute,

$$\begin{aligned} I_\eta &= \frac{1}{(2\pi i)^3} \int_{(3)} \int_{(3)} \int_{(2)} X^{1/2-z} Z^{x+z-1} Y^{w-1} \Phi(x+z-1) M(z) \widehat{W}_1(w) \\ &\quad \times \sum_{\substack{(6) \\ u}} q_u^{-z} \overline{\xi(u)} \mathcal{F}_{\eta,u}(x, w, z) dz dw dx. \end{aligned} \quad (5)$$

Here $\mathcal{F}_{\eta,u}$ factors as

$$\boxed{\mathcal{F}_{\eta,u}(x, w, z) = \frac{\zeta_F^S(6z) L_F^S(w, \psi_u)}{L_F^S(x, \eta \overline{\psi_u})} \mathcal{H}_{\eta,u}(x, w, z).} \quad (6)$$

The factor $\mathcal{H}_{\eta,u}$ is the explicit Euler product constructed in Section A.3. Appendix A proves (5)–(6) directly from (2), including all shared-prime cases and zero extensions. The row $u = 1$ gives the target denominator $L_F^S(x, \eta)$.

3 The low estimate: reflection and a quadratic mean square

The purpose of reflection is to replace the cubic coefficients by a sum with moving quadratic characters χ_R^3 . A quadratic large sieve can then control the mean square over the squarefree part R of the row index. We keep the remaining prime-power factors fixed while applying the sieve and sum them afterwards. This yields an average bound strong enough for the probe.

Use $V = V_G$ from Section 2, so $\widehat{V}(t) = e^{t^2}$. The dual variables introduced by reflection may meet S . For a fixed finite-order ray character ν supported on S , define

$$C_V(Z; m, \nu) = \sum_{\substack{c \text{ squarefree}, b \text{ primary} \\ (cb, S)=1}} \frac{\gamma_2(c) \overline{\alpha(cb^3)} \nu(cb^3) \chi_{cb^3}(m)}{q_c^{1/2} q_b} V\left(\frac{q_c q_b^3}{Z}\right). \quad (7)$$

There is no coprimality requirement between c and b . The Gaussian makes this series absolutely convergent.

After a fixed finite ray subdivision, the quadratic Hecke large sieve [3, Theorem 1.1 and Corollary 1.2] reads

$$\sum_{q_R \leq K}^{\text{sf}} \left| \sum_{q_n \leq N}^{\text{sf}} a_n \chi_R(n)^3 \right|^2 \ll_{\varepsilon} (KN)^{\varepsilon} (K+N) \sum_n |a_n|^2. \quad (8)$$

To match the cited theorem, use the primitive inducing quadratic characters. Their tame conductor is exactly R , and their remaining conductor divides a fixed modulus at 2, 3. A fixed ray subdivision makes these remaining local characters equal; their product then cancels, so coprime members in one class have primitive product conductor exactly their product. Quadratic reciprocity on these classes also reconciles the two orders of the residue symbol. These are the Hecke-family hypotheses of the theorem. Restricting back to the original support preserves every zero mask. Restricting the outer row set decreases the nonnegative left-hand side; a fixed coefficient-support restriction is imposed by setting the excluded coefficients to zero before applying the inequality.

Lemma 2 (Weak reflection). *Factor a nonzero row as $(m) = PsR$, where P is the product of the prime-power factors with exponent at least two, s is squarefree and supported on S , and R is squarefree and prime to SP . Freeze P, s and the unit of m . The completed row is a sum of $O_S(2^{\omega(P)})$ expressions of the form*

$$\zeta_R \sum_{\mu \neq 0} \frac{d(\mu) \alpha(\mu) \vartheta(\lambda^4 \mu)}{q_{\mu}^{1/2}} \chi_R(\lambda^4 \mu)^3 \prod_{p|D} B_p(\lambda^4 \mu) V^{\sharp} \left(\frac{q_{\mu} Z}{q_{c_F D R}^2} \right). \quad (9)$$

Here $\omega(P)$ counts the distinct prime divisors of P , $D \mid \text{rad}(P)$ is prime to S , c_F has finitely many fixed possibilities, and $|\zeta_R| \leq 1/81$. After a fixed finite ray subdivision of R , both d and ϑ are independent of R . The function $d(\mu)$ is the conjugate cusp coefficient defined in Appendix B; ϑ is its fixed additive phase. The frozen factors satisfy $|B_p(x)| \leq q_p^{1/2}$. The cusp support is

$$\mu = u \lambda^k n b^3, \quad k \geq -4, \quad n \text{ squarefree}, \quad |d(\mu)| \leq 27 3^{k/6} q_b^{1/2}.$$

The dual n, b are primary, may share primes, and need not avoid S . Every Euler derivative of $V^{\sharp}$ is bounded and decays faster than any power at infinity.

The proof, including the fixed cusp choices and the local zero masks, is given in Appendix B.

Lemma 3 (Balanced energy). *For every fixed $C > 0$, every fixed finite family of ν , and $\varepsilon > 0$, uniformly for $Z \geq 2$,*

$$\sum_{0 < q_m \leq CZ} |C_V(Z; m, \nu)|^2 \ll Z^{1+\varepsilon}.$$

Proof. Freeze P, s , a reflection branch and a ray sector. Put $p_0 = NP$, $d = ND \leq \sqrt{p_0}$, and place R in an annulus $q_R \asymp K \ll Z/p_0$. For a dual block $q_n \asymp N$, $q_b \asymp B$, fixed unit and $k \geq -4$, the coefficient bound is $d^{1/2}3^{-k/3}N^{-1/2}B^{-1}$. The remaining frozen coefficient is bounded by one and independent of R . Keep the kernel and the three annular cutoffs as one smooth profile in $(q_R/K, q_n/N, q_b/B)$. Simultaneous logarithmic Fourier inversion has a single common coefficient density with uniformly bounded L^1 norm. Indeed, in logarithmic coordinates its kernel is $V^\sharp(t_0 \exp(\xi_n + 3\xi_b - 2\xi_R))$, multiplied by three fixed compactly supported cutoffs. Its derivatives through order four are uniformly bounded for every $t_0 > 0$, by the Euler-derivative bounds. Applying $(1 - \Delta)^2$ bounds its Fourier transform by $C(1 + |t|^2)^{-2}$, which is integrable on $\mathbb{R}^3$. The constant is independent of P and every block parameter. Apply Minkowski to this common density before enlarging any positive row norm.

In each separated mode, freeze b first. The factors $\chi_R(u\lambda^{k+4})^3$ and $\chi_R(b)^3$ are then bounded row factors, including their zeros. Apply (8) in the squarefree n variable, separating its finitely many squarefree S -parts. With coefficients bounded by one, the squared norm is $\ll (KN)^\varepsilon(K + N)N$. Minkowski over the $O(B)$ choices of b and the displayed normalization give

$$\|\text{dual block}\|_2^2 \ll Z^\varepsilon d 3^{-2k/3} (K + N). \quad (10)$$

The kernel argument on the original support is comparable to $3^k NB^3 Z / (d^2 K^2)$. For arbitrarily small fixed $\tau > 0$, discard whole blocks with $3^k NB^3 > Z^\tau d^2 K^2 / Z$ by rapid decay. This is uniform: the cusp coefficient divided by $q_\mu^{1/2}$ is bounded, the frozen local factors and all row counts have a fixed polynomial cost, and lattice counting bounds the remaining absolute kernel tail. A sufficiently large fixed decay order therefore makes its total smaller than any prescribed power of Z . On the retained blocks insert $N \leq Z^\tau d^2 K^2 / (3^k Z B^3)$ into (10). Sum the geometric k factors and the logarithmically many annuli by the row-space triangle inequality. Reducing the preliminary losses gives, for fixed P ,

$$\sum_R |\text{branch}(R)|^2 \ll Z^\varepsilon d (K + d^2 K^2 / Z) \ll Z^{1+\varepsilon} \left(\frac{d}{p_0} + \frac{d^3}{p_0^2} \right) \ll Z^{1+\varepsilon} p_0^{-1/2}.$$

The reflection branches cost only a divisor factor by rowwise Cauchy. Every powerful ideal is uniquely $x^2 y^3$ with y squarefree, so ideal counting gives $\#\{P : NP \leq X\} \ll X^{1/2}$ and $\sum_{NP \leq CZ} (NP)^{-1/2} \ll \log(2Z)$. Summing over P and the finitely many remaining sectors proves the result. $\square$

Apply the exact separation (3) with $X = Y = Z^{1/2}$, so $Q_0 = q_{b*} Z$. The factor B is a fixed finite combination of completed rows (7); (4) gives squared additive norm $O(Z^{1/2})$, uniformly in the Mellin height. Cauchy, Lemma 3, and the integrability of $\widehat{W}_0(iv)$ now yield $|I_\eta(Z^{1/2}, Z^{1/2}, Z)| \ll Z^{-1/2} Z^{1/4} Z^{1/2+\varepsilon} = Z^{1/4+\varepsilon}$. Division by the fixed positive principal-residue normalizer c_S of Section 5 gives the same bound for J_η .

4 Buffered rows and their density

For the comparison estimate we must control many moving L -functions and their reciprocals at a finite height. We sort rows by the rightmost zero of their characters up to that height. Moving the contours a fixed distance to the right of each bin gives reciprocal bounds without crossing zeros. A bin farther to the right has a larger reflected-factor cost, but zero density bounds the number of such rows. The final estimate balances this cost against that count.

Fix the target η and a finite conjugation-stable family Θ of characters unramified outside S , containing $1, \eta, \bar{\eta}$. For a sixth-power-free element u prime to S , let $\mathcal{X}_u$ consist of the presentations

$$\psi_{u,\nu,\epsilon}(\mathfrak{n}) = \nu(\mathfrak{n})\psi_u(\mathfrak{n})^\epsilon, \quad \nu \in \Theta, \quad \epsilon \in \{1, -1\}.$$

All prescribed zero extensions are retained. If ψ^* is the primitive inducing character, then

$$L_{\text{orig}}(s, \psi) = L(s, \psi^*) \prod_{\mathfrak{p} \in E_\psi} (1 - \psi^*(\mathfrak{p})(N\mathfrak{p})^{-s}), \quad E_\psi \subset S \cup \{\mathfrak{p} : \mathfrak{p} \mid u\}.$$

Lemma 4 (Conductor and multiplicity). *For $U \leq Nu < 2U$, the primitive conductor has norm $O_{S,\Theta}(U)$. A given primitive character is induced by at most twelve physical rows. A nonunit row cannot induce the principal character.*

Proof. At $\mathfrak{p} \notin S$, put $q = N\mathfrak{p}$ and let $\lambda_{\mathfrak{p}}(x) = \bar{x}^{(q-1)/6} \in \mu_6$ on residue units. This character has exact order six. Tame local reciprocity identifies the restriction of the Kummer character of u to local units with $\lambda_{\mathfrak{p}}^{\kappa v_{\mathfrak{p}}(u)}$, where the fixed sign κ depends only on the Artin convention. The unit part of u does not enter this restriction. Twists in Θ are unramified there. Thus the conductor exponent is one precisely when $1 \leq v_{\mathfrak{p}}(u) \leq 5$, and zero otherwise.

The conductor at S is uniformly bounded: if $m > 2v_{\mathfrak{p}}(6)$, Hensel's lemma shows that $1 + \mathfrak{p}^m$ consists of sixth powers. Every local Kummer character is trivial there; the remaining twists have fixed conductors. The conductor therefore divides a fixed ideal times $\text{rad}(u)$.

For a fixed inducing character and orientation ϵ , its inertia restrictions determine each valuation $v_{\mathfrak{p}}(u) \in \{0, \dots, 5\}$ uniquely. There are no S -prime factors. Hence they determine at most one ideal (u) , which has six generators. The two orientations give the bound twelve. Nontrivial inertia at any prime of a nonunit proves the last assertion. $\square$

Lemma 5 (A zero-free disk). *Let $L(s, \psi)$ be primitive and nonprincipal, of conductor norm Q . Fix $0 < e < 10^{-3}$, $51/100 \leq a \leq 1$, and put $R_j = 2 - a - je$. If the disk $|s - (2 + it)| < R_2$ has no zero, then for every $\delta > 0$,*

$$|L(s, \psi)| + |L(s, \psi)^{-1}| \ll_{e,\delta} \{2Q(3 + |t|)^2\}^\delta \quad (|s - (2 + it)| \leq R_6).$$

The constants are uniform in a, Q, t, ψ .

Proof. For a primitive finite-order character the standard completion and functional equation are

$$\begin{aligned} \Lambda(s, \psi) &= (3Q)^{s/2} (2\pi)^{-s} \Gamma(s) L(s, \psi), \\ \Lambda(s, \psi) &= \epsilon_\psi \Lambda(1 - s, \bar{\psi}), \quad |\epsilon_\psi| = 1. \end{aligned}$$

This equation, Stirling's formula, and Euler convergence give the deliberately weak strip estimate $L(\sigma + iv, \psi) \ll Q^{3/5} (3 + |v|)^2$ for $-1/10 \leq \sigma \leq 11/10$; farther right use Euler convergence. Phragmen–Lindelöf follows from the usual theta integral, or by multiplying the normalized strip function by $e^{\delta(s-1/2)^2}$ and letting the height tend to infinity before $\delta \downarrow 0$.

Choose $g = \log L$ in the zero-free disk, normalized by the Euler logarithm at its center. Write $\mathcal{C} = 2Q(3 + |t|)^2$. The growth estimate and Borel–Carathéodory on radii R_3, R_4 give $\max_{R_4} |g| \ll_e \log \mathcal{C}$. On the inner disk of radius $r_0 = 49/100$, Euler convergence gives $|g| \ll 1$. Three circles therefore gives

$$\max_{R_6} |g| \ll_e (\log \mathcal{C})^{\theta_e}, \quad \frac{\log(R_6/r_0)}{\log(R_4/r_0)} \leq \theta_e < 1.$$

The strict uniform inequality follows by compactness in a . Exponentiating g and $-g$, and using $\exp(C_e (\log \mathcal{C})^{\theta_e}) \ll_{e,\delta} \mathcal{C}^\delta$, proves the result. $\square$

Deleted factors and their inverses cost at most $(N \text{rad}(u))^\delta$ on any fixed half-plane $\text{Re } s \geq \sigma_0 > 0$. Indeed each logarithm is $O_{\sigma_0}((N \mathfrak{p})^{-\sigma_0})$, which is at most $\delta \log N \mathfrak{p}$ outside a fixed finite set. On $\text{Re } s \geq -v$, only the direct product is needed, and it costs $(N \text{rad}(u))^{v+\delta}$.

Global reciprocal bounds. For a principal denominator apply the same disk proof to $(s-1)\zeta_F(s)/(s+1)$ and then use $1/\zeta_F = (s-1)/(s+1) \cdot [(s-1)\zeta_F/(s+1)]^{-1}$. The regularized value at one is nonzero. Consequently all reciprocals have the corresponding all-height bounds on any fixed line strictly to the right of the family zero supremum β_* .

Lemma 6 (Fixed-height bins). *Assume $\beta_* \geq 51/100$, as in the later contradiction argument. For each nonunit row of norm comparable to U , and any $T \geq 3$, one can assign one of $O_e(1)$ labels $a \in [51/100, \beta_*]$ such that:*

1. *If $a > 51/100$, some inducing function in $\mathcal{X}_u$ has a zero with real part at least a and imaginary part of absolute value at most $2T$.*
2. *For every presentation in $\mathcal{X}_u$,*

$$|L_{\text{orig}}(\sigma + it)| + |L_{\text{orig}}(\sigma + it)^{-1}| \ll U^\delta (3+T)^\delta \quad (\sigma \geq a + 6e, |t| \leq T).$$

3. *On the reflected line,*

$$|L_{\text{orig}}(1 - a - 6e + it)| \ll U^{a-1/2+12e+\delta} (3+T)^A \quad (|t| \leq T),$$

where A is fixed.

Proof. Let M_u be the maximum of $51/100$ and the real parts of all zeros of the finite inducing family with $\text{Re } \rho \geq 51/100$ and $|\Im \rho| \leq 2T$. Put $a = 51/100 + e[(M_u - 51/100)/e]$. Euler convergence bounds these real parts by one, so the maximum exists by discreteness of zeros in a compact rectangle. If the label is above the floor it supplies an actual zero.

For $|t| \leq T$, the disk centered at $2 + it$ of radius $2 - a - 2e$ has real part greater than M_u and height less than $T + 3/2 < 2T$. Apply Lemma 5, the conductor bound, and the deleted-factor estimate. Shrinking preliminary losses gives the stated direct bound. The functional equation reflects the other line to $a + 6e - it$ for the conjugate primitive character. It costs $Q^{a-1/2+6e}$ and a fixed height power. Its deleted factors cost at most $U^{6e+\delta}$. This proves the last bound. $\square$

The published general Hecke zero-density theorem [4, Lemma 3.3], restricted to angular type zero over our degree-two field, states, uniformly for $1/2 \leq \sigma < 1$, $Q > 1$, and $T > 2$,

$$\sum_{N\mathfrak{f} \leq Q} \sum_{\psi \bmod \mathfrak{f}}^* N(\sigma, T, \psi) \ll (Q^2 T^2)^{h(\sigma)(1-\sigma)} (\log QT)^B, \quad h(\sigma) = \min \left\{ \frac{3}{2-\sigma}, \frac{2}{\sigma} \right\} \leq \frac{5}{2}.$$

Here N counts zeros with real part at least σ and height at most T , and the star means primitive. Lemmas 4 and 6, followed by the trivial lattice count, imply

$$\#\mathcal{B}_a \ll U^{R(a)+\delta} (3+T)^A, \quad R(a) = \min\{1, 5(1-a)\}. \quad (11)$$

The floor uses only the trivial count. At a possible label $a = 1$, take $\sigma \uparrow 1$ in the uniform theorem for fixed conductor and height ranges.

5 The nonvanishing numerator and the principal row

The signal comes from the unique row whose numerator character is principal. We first identify that row by local character calculations and the Chinese remainder theorem. We then show that its Euler correction H_η cannot vanish in the required half-plane: a zero of this correction could otherwise cancel a pole of $1/L_F^S$.

Lemma 4 excludes nonunit rows with principal numerator. To exclude the five nonidentity units, take the element $\pi = -2 - 3\omega$, which is primary and has norm seven. By CRT choose

$$n \equiv \pi \pmod{36\mathcal{O}}, \quad n \equiv 1 \pmod{\mathfrak{p}} \quad (\mathfrak{p} \in S, \mathfrak{p} \nmid 6).$$

Then n is primary, prime to S , and $Nn \equiv 7 \pmod{36}$. For a unit $u \in \mu_6$, the defining symbol gives $\chi_{\mathfrak{p}}(u) = u^{(N\mathfrak{p}-1)/6}$. Since every good prime norm is 1 modulo six, multiplication modulo 36 yields $\chi_n(u) = u^{(Nn-1)/6} = u$. Thus each of the five nonidentity units has a nonprincipal numerator, even after an arbitrary finite enlargement of S .

Thus $u = 1$ is the only principal numerator row. To prove positivity of $M(1/6)$, use, for $0 < \sigma < 1$,

$$|y|^{2\sigma-2} = \frac{1}{\Gamma(1-\sigma)} \int_0^\infty t^{-\sigma} e^{-t|y|^2} dt.$$

Pair this identity with the Schwartz function $\widetilde{W}_0(|y|^2)$. The pairing is absolutely convergent. Fourier duality turns each Gaussian into a positive Gaussian; its pairing with the nonnegative, nonzero annular $W_0(|x|^2)$ is strictly positive. The resulting integral is absolutely convergent also on this side, by annular support. Polar coordinates identify the original pairing with a positive constant times $M(\sigma)$. Hence $M(1/6) > 0$. Writing $\rho_S = \text{Res}_{s=1} \zeta_F^S(s) > 0$, we obtain $c_S = \widehat{W}_1(1)M(1/6)\rho_S^2/6 > 0$. Both zeta residues contribute to this constant. We use the same normalized physical probe $J_\eta = I_\eta/c_S$ in both representations.

At the principal residue define $H_\eta(s) = \mathcal{H}_{\eta,1}(s, 1, 1/6)$. With $t = Q^{-1}$, $D = \eta(\mathfrak{p})Q^{-s}$, and $R = \overline{\alpha(\mathfrak{p})}^6 \eta(\mathfrak{p})^6 Q^{3-6s}$, the local factor simplifies to

$$H_{\mathfrak{p}}(s) = \frac{1-t^2}{1-R} \left(1 + \frac{t(D-R)}{1-D} \right). \quad (12)$$

Consequently on $\text{Re } s \geq 11/12$, uniformly in heights and unit phases, $H_{\mathfrak{p}}(s) - 1 \ll Q^{-2} + Q^{-1-\text{Re } s} + Q^{3-6\text{Re } s}$. These exponents have a fixed prime-summability margin, and the two denominators in (12) stay away from zero. After putting all sufficiently small primes into S , the product is holomorphic and satisfies

$$|H_\eta(s) - 1| \leq \frac{1}{2} \quad (\text{Re } s > 11/12). \quad (13)$$

The cutoff can be chosen uniformly in the target's unit phases; additional target conductor primes may then be deleted.

6 The finite-height contour comparison

We now extract the signal and bound the remaining rows. Two residues in the principal row produce f_η . For nonprincipal rows we use the buffered bounds and row counts from Section 4. The contours are truncated before using these finite-height bounds; rapid decay controls the discarded tails. Large rows are estimated on a separate contour where absolute convergence suffices.

Set $X = Y = Z^{1/2}$ in (5)–(6), and write $I_\eta(Z) = I_\eta(Z^{1/2}, Z^{1/2}, Z)$. We use the normalization $J_\eta = I_\eta/c_S$ and correction H_η from Section 5. The physical probe is independent of the contour height chosen below. The principal signal is

$$f_\eta(Z) = \frac{1}{2\pi i} \int_{(2)} Z^{x-2/3} e^{(x-5/6)^2} \frac{H_\eta(x)}{L_F^S(x, \eta)} dx. \quad (14)$$

Local domains. Write $x_r = \operatorname{Re} x$, $w_r = \operatorname{Re} w$, and $z_r = \operatorname{Re} z$. Here is the local verification needed for all contour paths. For a good prime of norm Q , put

$$d = \eta(\mathfrak{p})Q^{-x}, \quad h = Q^{-w}, \quad E = dh, \quad V = Q^{-6z}, \quad R = a_{\mathfrak{p}}^2 Q^{4-6x-6z},$$

where $a_{\mathfrak{p}}$ is the modulus-one arithmetic phase defined in Section A.3. At a prime $\mathfrak{p} \nmid u$, $\rho = \chi_{\mathfrak{p}}(u)$ satisfies $\rho^6 = 1$, and the finite Euler identity is

$$H_{\mathfrak{p}}(\rho) = \frac{1}{1-R} \left(1 + \frac{B_0 + dV\rho^{-1} + A_1\rho + A_2\rho^2}{1-d\rho^{-1}} \right), \quad (15)$$

$$B_0 = E(1-QV) - R/Q, \quad A_1 = h[-V + R/Q + E(Q-1)V], \quad A_2 = -R(1-V)h^2.$$

For $j = v_{\mathfrak{p}}(u) \geq 1$, reset $\rho = \chi_{\mathfrak{p}}(u/\pi_{\mathfrak{p}}^j)$ for the fixed primary generator $\pi_{\mathfrak{p}}$, so that $\rho^6 = 1$; the full ramified factors are

$$H_{\mathfrak{p},1} = \frac{1 - R/Q + E(1-QV)}{1-R},$$

$$H_{\mathfrak{p},j} = \frac{1 - R/Q - E(Q-1) + (1-V)J_j}{1-R} \quad (j \geq 2), \quad (16)$$

Here $J_2, \dots, J_5$ are given in (A.3), with $|a_{\mathfrak{p}}| = |b_{\mathfrak{p}}| = 1$. The term $-E(Q-1)$ occurs for every $j \geq 2$, since the local formula contains $V^{\mathbf{1}_{j \leq 1}}$.

On the domain

$$x_r \geq \frac{51}{100}, \quad z_r \geq \frac{17}{50}, \quad w_r \geq -\frac{1}{100}, \quad x_r + w_r \geq 1 + \delta, \quad \delta > 0, \quad (17)$$

we have $|d| < 1$ and $|R| \leq Q^{-11/10}$. In (15), the unit monomials have decay exponent greater than one: E has exponent at least $1 + \delta$, and the other monomials have a fixed margin. For example the exponents of $hV, hEQV, Rh^2$ are at least $203/100, 203/100 + \delta, 52/25 + 2\delta$. In (16), the additional $E(Q-1)$ has exponent at least δ , and each J_j has exponent at least $1/50$. Thus normal convergence at the unramified primes and the elementary divisor bound at the ramified primes prove

$$\mathcal{H}_{\eta,u} \ll_{\delta,\epsilon} q_u^\epsilon \quad \text{on every fixed real box in (17),} \quad (18)$$

uniformly in all heights.

We also use the larger principal box

$$x_r \geq \frac{15}{16}, \quad w_r \geq \frac{3}{4}, \quad z_r \geq \frac{1}{8}. \quad (19)$$

Here $|d| \leq Q^{-15/16}$ and $|R| \leq Q^{-19/8}$. All unit monomials in (15) have exponent at least $23/16$; the limiting one is EQV . Consequently $\mathcal{H}_{\eta,1} \ll 1$ on this whole box. The unramified decay exponents increase with x_r, w_r, z_r , so the bound holds throughout the principal contour paths. For the separate large-row domain $x_r, w_r, z_r \geq 2$, the ramified term $E(Q-1)$ has decay exponent at least three, and every J_j has exponent greater than one. Thus $\mathcal{H}_{\eta,u} \ll 1$ uniformly for all rows on that domain.

A common tail bound. On fixed real strips, $\widehat{W}_1$ is Schwartz, and M is holomorphic and Schwartz on compact positive real strips. For M , integrate by parts in the logarithmic radial variable; the endpoint at zero is integrable because $z_r > 0$. The independent height coordinates are $(t_x + t_z, t_z, t_w)$. Hence for every L ,

$$|e^{(x+z-1)^2} M(z) \widehat{W}_1(w)| \ll_L (1 + |t_x| + |t_z| + |t_w|)^{-L}.$$

If the arithmetic factor on a specified domain is bounded by $Z^B(1 + |\mathbf{t}|)^A$, integration outside a height box, or on any horizontal face, is $O_N(Z^B T^{-N})$: take $L > N + A + 3$. The same proof applies to the two variables remaining after the w residue. Increasing N changes only external test seminorms, not A or B . All reciprocal bounds are used only inside their stated domains.

For global reciprocal estimates, including principal denominators, we use the regularization established in Section 4, under “Global reciprocal bounds.”

Nonunit rows below the large-row cutoff. For $Z \geq 2$, fix a dyad $U = Z^d$, $0 \leq d \leq 501/1000$, and a buffered bin $\mathcal{B}$ of nonunit rows of label a before moving any variable. Set $z_0 = 17/50$ and choose a fixed small $e > 0$. Starting in (5)–(6), move z to z_0 and x to $\beta_* + 20e$, with $w = 3$; then move w to $1 - a - 6e$. Throughout, $x_r + w_r \geq 1 + 14e$, the numerator is nonprincipal, the reciprocal remains to the right of β_* , and $6z_r > 1$. The paths lie in (17), so no pole is crossed and the tail bound justifies the infinite horizontal limits.

While x is still global, discard the complement of $|t_x|, |t_w|, |t_z| \leq T/4$. Its cost is $O_N(Z^B T^{-N})$. Move only the retained x segment to $a + 16e$. The buffered rectangle contains this segment and its two horizontal joins; $x_r + w_r = 1 + 10e$. On those joins the w, z axes may be extended using global numerator bounds, because the reciprocal depends only on the still-buffered x . The face bound again gives $O_N(Z^B T^{-N})$. No moved reciprocal axis is extended beyond its zero-free rectangle.

On the retained contour the buffered functional equation gives $L_F^S(w, \psi_u) \ll U^{a-1/2+12e+\epsilon}(1+T)^A$. The outside real power is $Z^{a/2-3/4+17/100+13e}$. Using (18), allocating the small powers, and then taking e sufficiently small proves

$$|E_{d,a}(Z, T)| \ll_{\eta, \epsilon} Z^{a/2-3/4+17/100+\epsilon} U^{a-21/25} \#\mathcal{B} (1+T)^A. \quad (20)$$

The principal row. Suppose $\beta_* > 47/48$. Move its contours to $x_r = \beta_* + e$, $w_r = 1 + e$, $z_r = 1/6 + e$. Next move w to $3/4$, crossing only the simple pole at $w = 1$. In that residue move z to $1/8$, crossing only the simple pole at $z = 1/6$. The unresidued w integral keeps $z_r = 1/6 + e$. Every path lies in (19); M has no pole there, and the reciprocal stays global. The two-variable tail bound justifies the second move after the first residue. The double residue is exactly $c_S f_\eta(Z)$, after moving its remaining x line right to 2. No pole is crossed in this last move, including for principal η , whose reciprocal has a zero at 1. The two other integrals satisfy

$$\begin{aligned} E_w &\ll_\eta Z^{\beta_*-2/3-1/8+(3/2)e+\epsilon}, \\ E_z &\ll_\eta Z^{\beta_*-2/3-1/48+e+\epsilon}. \end{aligned} \quad (21)$$

These follow directly from the balanced outside exponent $x_r + z_r/2 + w_r/2 - 5/4$. Both surviving principal remainders have a fixed margin.

The remaining unit rows. For each of the five units $u \neq 1$, the numerator is nonprincipal by Section 5. Move to $(x_r, w_r, z_r) = (\beta_* + e, 1/2, 17/50)$ through (17). No pole is crossed; the reciprocal stays to the right of β_* . The fixed-conductor numerator has polynomial height growth, so the

common tail bound and the outside exponent give

$$\sum_{u \in \mu_6 \setminus \{1\}} |E_u| \ll_{\eta} Z^{\beta_* - 2/3 - 49/300 + e + \epsilon}, \quad \frac{17}{100} + \frac{1}{4} - \frac{5}{4} = -\frac{2}{3} - \frac{49}{300}. \quad (22)$$

Large nonunit rows. For $U > Z^{501/1000}$ move each dyad to $(2, 2, v)$ with fixed $v > 2$. The path stays in $x_r, w_r, z_r \geq 2$, where both L -arguments are absolute and the all-row Euler bound is uniform. Thus

$$|E_U| \ll Z^{7/4 + v/2 + \epsilon} U^{1 - v + \epsilon}, \quad \sum_{U > Z^{501/1000}} |E_U| \ll Z^{7/4 + 501/1000 - v/1000 + \epsilon'}. \quad (23)$$

With $v = 4000$ this is $O(Z^{-1})$. Initial absolute convergence and the summable final bound justify summing the separate dyadic contour identities.

Equations (21)–(23), after division by c_S , give a principal, unit, and large-row remainder

$$O_{\eta, \epsilon, N}(Z^{C(\beta_*) - m_0 + \epsilon}(1 + T)^A + Z^B T^{-N}).$$

The exponents A, B are independent of N ; we choose the losses, T , and N in the next proof.

7 The comparison estimate and the choice of constants

The numerical choices leave room for contour buffers, height factors, and logarithmic losses. Two basic margins explain the choice $b = 47/48$:

$$b - \frac{29}{30} = \frac{1}{80}, \quad b - \frac{2}{3} - \frac{1}{4} = \frac{1}{16}.$$

The first is the margin over the worst nonunit-row exponent computed below; the second makes the low estimate smaller than the signal's allowed growth. The principal-row errors and the domain of H_{η} must also be respected, as verified in the preceding sections.

The cutoff $0.501 = 1/2 + 1/1000$ extends the nonunit-row estimate slightly beyond the balanced scale. Its extra exponent cost is at most $1/1000$, less than the first margin above; the positive excess over $1/2$ also allows a sufficiently far contour to control larger rows. We reserve $m_0 = 1/1200$, smaller than the available margins, and keep only $\delta_0 = m_0/4 = 1/4800$ after absorbing the remaining losses. These values provide convenient safety margins; the argument needs positive margins, not these exact fractions.

Lemma 7. *Suppose $\beta_* > b$. For $Z \geq 2$, $|J_{\eta}(Z) - f_{\eta}(Z)| \ll_{\eta} Z^{C(\beta_*) - 1/4800}$. The saving $1/4800$ is independent of the target.*

Proof. Insert (11) in (20). Before the adjustable losses and height factor, the exponent relative to $Z^{C(\beta_*)}$ is

$$E(d, a) = \frac{a}{2} - \beta_* + \frac{13}{150} + d \left(R(a) + a - \frac{21}{25} \right).$$

At $d = 1/2$, this is $a - \beta_* + R(a)/2 - 1/3$. The two cases give

$$a + \frac{R(a)}{2} - \frac{1}{3} \leq \begin{cases} 4/5 + 1/2 - 1/3 = 29/30, & a \leq 4/5, \\ 13/6 - 3a/2 \leq 29/30, & a \geq 4/5. \end{cases}$$

Since $\beta_* > 47/48$, we obtain $E(1/2, a) < -1/80$. Also $0 < R(a) + a - 21/25 \leq 1$ throughout $51/100 \leq a \leq 1$. Thus $E(d, a)$ increases with d , and throughout $0 \leq d \leq 501/1000$,

$$E(d, a) \leq E(1/2, a) + \frac{1}{1000} < -\frac{1}{80} + \frac{1}{1000} = -\frac{23}{2000} < -\frac{1}{100}.$$

This saving for all nonunit dyads below the cutoff is larger than m_0 . Choose all preliminary losses and absorb the logarithmic number of terms so that the total real loss is at most $m_0/4$. The row count and contour comparison now give, for every fixed N ,

$$|J_\eta - f_\eta| \ll_{\eta, N} Z^{C(\beta_*) - 3m_0/4} (1+T)^A + Z^B T^{-N}.$$

After fixing the target, choose $0 < \tau \leq m_0/[4(A+1)]$ and $T = 3Z^\tau$, so the hypothesis $T \geq 3$ in Lemma 6 always holds. The first term is $O_\eta(Z^{C(\beta_*) - m_0/2})$. Choose N so large that $B - N\tau < C(b) - m_0/4$. The second term is then $O_\eta(Z^{C(\beta_*) - m_0/4})$. The external-tail argument permits this late choice without changing the previously fixed orders. Since $m_0/4 = 1/4800$, the result follows. $\square$

8 Mellin continuation excludes the zeros

We prove the Hecke assertion of Theorem 1. Suppose $\beta_* > b$. The proved balanced low estimate with $\varepsilon = 1/32$ gives $J_\eta(Z) \ll_\eta Z^{9/32}$. As

$$C(\beta_*) - \frac{9}{32} > C(b) - \frac{9}{32} = \frac{1}{32},$$

Lemma 7 and the triangle inequality imply the following bound for $Z \geq 2$. The defining integral makes f_η continuous on $[1, 2]$, so enlarging the implied constant extends it to $Z \geq 1$:

$$f_\eta(Z) \ll_\eta Z^{C(\beta_*) - 1/4800} \quad (Z \geq 1). \quad (24)$$

For $0 < Z \leq 1$, the contour in (14) can be moved to any fixed line $\operatorname{Re} s = M > 2$. Indeed, the reciprocal Euler product and H_η are bounded there, and the Gaussian decays rapidly on vertical lines and horizontal joins in a fixed strip. Thus

$$f_\eta(Z) \ll_{\eta, M} Z^{M-2/3} \quad (0 < Z \leq 1).$$

Together with (24), this makes

$$F_\eta(s) = \int_0^\infty f_\eta(Z) Z^{-s+2/3} \frac{dZ}{Z}$$

locally uniformly convergent, and hence holomorphic, on $\operatorname{Re} s > \beta_* - 1/4800$.

To identify the transform, put $A_\eta(s) = e^{(s-5/6)^2} H_\eta(s)/L_F^S(s, \eta)$ for $\operatorname{Re} s > 1$. Writing $Z = e^u$, the defining integral gives

$$e^{-4u/3} f_\eta(e^u) = \frac{1}{2\pi} \int_{\mathbb{R}} e^{itu} A_\eta(2+it) dt.$$

The right-hand Fourier density is continuous and integrable. The left-hand function is integrable by the two endpoint bounds just proved. Fourier inversion gives $F_\eta(2+it) = A_\eta(2+it)$; the identity theorem therefore identifies them on $\operatorname{Re} s > 1$.

Since $\beta_* - 1/4800 > 11/12$ and $|H_\eta| \geq 1/2$, the function $e^{-(s-5/6)^2} F_\eta(s)/H_\eta(s)$ continues $1/L_F^S(s, \eta)$ holomorphically to $\operatorname{Re} s > \beta_* - 1/4800$. The same positive distance works for every target, although its constants and height choice may differ. By the definition of β_* , some primitive target has a zero with real part greater than $\beta_* - 1/4800$. Its reciprocal has a pole there, which contradicts this continuation. Thus $\beta_* \leq 47/48$.

An imprimitive Hecke function differs from its primitive inducing function by finitely many Euler factors, all nonzero on $\operatorname{Re} s > 0$. This proves the assertion for all finite-order Hecke characters.

9 Transfer to Dirichlet functions

Let χ be a primitive Dirichlet character, let χ_{-3} be the quadratic character modulo 3, and pull χ back to ideals by the norm map. This defines a finite-order Hecke character η . After deleting the finitely many Euler factors above primes dividing $3 \operatorname{cond}(\chi)$, one has

$$L_F^S(s, \eta) = L^{S_{\mathbb{Q}}}(s, \chi) L^{S_{\mathbb{Q}}}(s, \chi \chi_{-3}).$$

At a split rational prime the two norm- p factors on the left coincide; at an inert prime the norm- p^2 factor is

$$(1 - \chi(p)^2 p^{-2s})^{-1} = (1 - \chi(p) p^{-s})^{-1} (1 + \chi(p) p^{-s})^{-1}.$$

These prove the identity first for $\operatorname{Re} s > 1$, and then by meromorphic continuation. In $47/48 < \operatorname{Re} s < 1$, both Dirichlet factors are holomorphic, so a zero of either would contradict the Hecke conclusion. The same reasoning applies at $1 + it$ for $t \neq 0$; absolute Euler convergence handles $\operatorname{Re} s > 1$.

At $s = 1$ a pole-zero cancellation could only involve a principal inducing character and χ_{-3} . But

$$L(1, \chi_{-3}) = \sum_{n \geq 0} \left(\frac{1}{3n+1} - \frac{1}{3n+2} \right) > 0.$$

The series is convergent, and grouping the ordinary convergent Dirichlet series into these adjacent pairs is valid. Thus there is no such cancellation. Deleted factors are nonzero in the region, and the same observation handles imprimitive Dirichlet characters. This completes the deduction of Theorem 1.

A Arithmetic identities for the probe

This appendix proves the exact high representation (5)–(6) for the physical probe (2), with the definitions of Section 2.

A.1 Gauss sums and the fixed phase correction

The function Γ_q in (1) is the normalized quadratic Gauss sum: $\Gamma_q(c) = |c|^{-1} \sum_{x \bmod c} e(x^2/c)$. One short verification applies Poisson summation to $e(z^2/c) e^{-\pi \delta |z|^2}$: its transform is

$$\frac{|c|}{2\sqrt{r}} e^{-\pi \delta q_c |y|^2 / (4r)} e(-cy^2 / (4r)), \quad r = 1 + 3q_c \delta^2 / 16.$$

Divide by $2/(\sqrt{3}\delta)$ and let $\delta \downarrow 0$. The two periodic Gaussian means give

$$\Gamma_q(c) = \frac{1}{2} \sum_{y \bmod 2\mathcal{O}} e(-cy^2/4),$$

which is (1). The replacement of r by one has error $O(1)$ before this division, by the Gaussian second-moment bound.

In the quotient of $(\mathcal{O}/4\mathcal{O})^\times$ by its subgroup of squares, the four classes $1, -1, \lambda, -\lambda$ have Γ_q -values $1, 1, i, -i$. Thus R is a symmetric sign-valued bicharacter; explicitly its value on $(-1)^e \lambda^f, (-1)^g \lambda^h$ is $(-1)^{eh+fg+fh}$. Cubic reciprocity gives $\chi_c(4) = (c/(-2))_3$, a character modulo 2. Consequently

$$G(vw) = G(v)G(w)R(v, w), \quad G(p^3) = \gamma_3(p), \quad R(p, p) = \gamma_3(p)^2 = \chi_p(-1). \quad (\text{A.1})$$

This also proves the fixed-ray-group assertion in Section 2. The prime Gauss–Jacobi identities, with these phases, are

$$\gamma_2(p)^3 = -\alpha(p), \quad \gamma_1(p)\gamma_2(p) = -\alpha(p)G(p).$$

Indeed, write $Q = Np$ and $J(\psi, \psi) = \sum_{x \in \mathcal{O}/p} \psi(x)\psi(1-x)$, with $\psi = \chi_p^2$. Reduction modulo p makes the summand $x^{(Q-1)/3}(1-x)^{(Q-1)/3}$; its degree is less than $Q-1$, so the finite-field sum is zero. Thus $p \mid J$. The Gauss–Jacobi identity gives $|J| = \sqrt{Q}$. On $\mathbb{F}_Q \setminus \{0, 1\}$, the order-three map $x \mapsto 1/(1-x)$ preserves the summand. Its two fixed points satisfy $x(1-x) = 1$, and all other orbits have size three. Hence $J \equiv 2 \equiv -1 \pmod{3}$. The primary convention therefore gives $J = -p$. For $\chi = \chi_p$, the substitutions $x = (1+y)/2$ and $y \mapsto y/(y-1)$ in the Jacobi sums give

$$J(\chi^2, \chi^2) = \overline{\chi(4)}^2 J(\chi^3, \chi^2) = \overline{\chi(4)}^2 J(\chi, \chi^2).$$

Since $\chi(4)^3 = 1$, this and $g_\chi g_{\chi^2} = J(\chi, \chi^2)g_{\chi^3}$ give the second formula. Writing $\mu(c)$ for the ideal Möbius function, CRT then gives

$$\gamma_2(c) = \frac{\mu(c)\alpha(c)G(c)}{\gamma_1(c)}, \quad \gamma_1(c) = \prod_{p|c} \gamma_1(p) \prod_{p < r, p, r|c} \chi_p(r)\chi_r(p). \quad (\text{A.2})$$

Thus the correction is an explicit fixed function, also on nonsquarefree indices; it is never interpreted as a nonsquarefree Gauss sum.

A.2 Poisson summation and the moving-prime phases

For a nonzero sixth-power-free u prime to S , including its unit, put

$$\begin{aligned} F(s, A, H) &= \sum_{\substack{d \bmod s \\ H \equiv b_* A d \pmod{s}}} \chi_s(d) g_{\xi \chi_A} \left(b_* A, \frac{H - b_* A d}{s} \right), \\ K_\eta(c, n, s, a; u) &= \frac{F(s, cn^3, ua^6) \chi_s(b_*)}{\sqrt{q_{b_*}} \tau \xi(s) \xi(u)} \gamma_2(c) \alpha(cn^3) \overline{G(cn^3)} \eta(cn^3) \Xi(cn^3)^{-1} R(cn^3, s), \\ \mathcal{F}_{\eta, u}(x, w, z) &= \sum_{c \text{ sf}, n, s, a} K_\eta(c, n, s, a; u) q_c^{-1/2-x} q_n^{-1-3x} q_s^{-w} q_a^{-6z}. \end{aligned}$$

To prove (5), use the full modulus sb_*A , even when s, A share primes. Write a residue as $r + b_*A\ell$ and expand the s -Gauss sum. The ℓ sum imposes $H \equiv b_*A d \pmod{s}$, giving finite transform $q_s^{1/2} F(s, A, H)$. The planar dilation contributes X/q_A ; including the outer square root leaves $X^{1/2}/(\sqrt{q_{b_*}} q_A)$. The radial Fourier argument is $X q_H/q_A$. Primitivity of ξ removes all frequencies meeting S , including zero. Write uniquely $H = ua^6$, with a primary and u retaining the unit of H ; thus this change of variables introduces no factor six. Mellin inversion and $x = t + 1 - z$ now give every factor in (5). The bound $|F| \leq q_s q_{b_*} q_A$ verifies absolute convergence on these lines.

Fixed and moving phase cancellation. CRT gives

$$g_{\xi\chi_A}(b_*A, K) = \xi(A)\chi_A(b_*)g_{\xi}(b_*, K)g_{\chi_A}(A, K).$$

At S , $K = (H - b_*Ad)/s \equiv H/s$, so its factor is $\sqrt{q_{b_*}}\tau\xi(A)\overline{\xi(u)}\xi(s)$. The denominator in K_η and $\Xi(A)^{-1}$ cancel all ξ factors. The remaining b_* symbols cancel as $\prod_p \chi_p(b_*)^{t_p-k_p} = \chi_A(b_*)/\chi_s(b_*)$. Here $t_p = v_p(A)$ and $k_p = v_p(s)$.

For the other CRT units write $h_p = H/p^{v_p(H)}$, $s_p = s/p^{k_p}$, and $A_p = A/p^{t_p}$. Their factor is

$$U_p = \chi_p(h_p)^{k_p-t_p}\chi_p(s_p)^{t_p}\chi_p(A_p)^{t_p-k_p}\chi_p(b_*)^{t_p-k_p}.$$

Reciprocity cancels the off-diagonal A, s phases exactly:

$$R(A, s) \prod_p \chi_p(s_p)^{t_p}\chi_p(A_p)^{-k_p} = \prod_p R(p, p)^{t_p k_p}.$$

Writing $t_p = e_p + 3l_p$ with $e_p = v_p(c)$, division by the pair factors of $\gamma_1(c)$ leaves a product over $p < r$, with factor $R(p, r)^{e_p l_r + e_r l_p + l_p l_r}$. The refinement law (A.1) expresses that product as

$$R(c, n)G(n^3) \prod_p \overline{G(p^3)}^{l_p} R(p, p)^{-e_p l_p - l_p(l_p-1)/2}.$$

Its first two factors combine with $G(c)$ to cancel the inserted $\overline{G(cn^3)}$. Also $\alpha(c)\overline{\alpha(cn^3)} = \overline{\alpha(n)}^3$. These identities hold even when c, n share primes.

A.3 Local Euler factors

For $p \notin S$ set $Q = Np$, $j = v_p(u)$, $\rho = \chi_p(u/p^j)$, $\omega_p = \chi_p(-1)$, and

$$a_p = \overline{\alpha(p)}^3 \eta(p)^3, \quad b_p = \overline{\alpha(p)}^2 \eta(p)^2 \overline{\chi_p(4)}, \quad V = Q^{-6z}, \quad \mathcal{R} = a_p^2 Q^{4-6x-6z}, \quad W_\ell = \rho Q^{-w}.$$

Thus $b_p^3 = a_p^2$. The factor of $\mathcal{F}_{\eta, u}$ at p is

$$P_p^* = \frac{1}{1-\mathcal{R}} \left\{ \frac{\mathcal{R}(1-Q^{-1}) - \eta(p)(Q-1)Q^{-x-w}V^{\mathbf{1}_{j \leq 1}}}{1-V} + J_j \right\},$$

$$P_p = \frac{1}{1-V} + \mathbf{1}_{j=0} \frac{W_\ell}{1-W_\ell} + P_p^*,$$

where

j	J_j
0	$-\eta(p)\rho^{-1}Q^{-x} + W_\ell \mathcal{R}$
1	$\eta(p)Q^{-x-w}$
2	$a_p \rho^{-3} Q^{3/2-3x}$
3	$-\eta(p)b_p \rho^{-2} Q^{2-3x-w} + b_p^2 \rho^{-4} Q^{2-4x}$
4	$-\eta(p)a_p \rho^{-3} Q^{5/2-4x-w}$
5	$-a_p^2 Q^{3-6x}.$

(A.3)

Here is a direct finite verification of the table. For $t \geq 1$, summing over lifts of a residue modulo p gives

$$g_{\chi_p^h}(p^t, K) = \begin{cases} Q^{t-1} g_{\chi_p^h}(p, K/p^{t-1}), & p^{t-1} \mid K, \\ 0, & p^{t-1} \nmid K. \end{cases}$$

At prime modulus, a unit frequency κ contributes $\chi_p(\kappa)^{-h} g_{\chi_p^h}(p, 1)$; a zero frequency contributes 0 for $h \not\equiv 0 \pmod{6}$ and $Q - 1$ otherwise. In the latter case the unit-frequency sum is -1 . Before restoring the units, the local convolution is

$$C_p(t, k, j') = \sum_{\substack{d \bmod p^k \\ p^{j'} \equiv p^t d \pmod{p^k}}} \chi_p(d)^k g_{\chi_p^t} \left(p^t, \frac{p^{j'} - p^t d}{p^k} \right).$$

When $k = 0$, the absent character factor is defined to be one; when $t = 0$, the modulus-one Gauss factor is one. These empty factors are distinct from zeroth powers at positive modulus, which retain their zero values at nonunits. It equals 1 for $t = k = 0$, $\mathbf{1}_{j'=0}$ for $t = 0 < k$, $g_{\chi_p^t}(p^t, p^{j'})$ for $k = 0 < t$, and $\omega_p^{-1} G_1 g_{\chi_p^{t-1}}(p^t, p^{j'-1})$ for $t > 0, k = 1, j' \geq 1$, where $G_1 = g_{\chi_p}(p, 1)$. All other cases vanish. For $t > 0$ and $k \geq 2$, write $d = d_0 + p^{k-1}h$. The congruence and character are unchanged, while expanding the inner Gauss sum at a unit y produces $\sum_{h \bmod p} e(-hy/p) = 0$. The restored summand, with $t = e + 3l$, is

$$(-1)^e \gamma_1(p)^{-e} \eta(p)^e (a_p \overline{\gamma_3(p)})^l \rho^{k-t} \omega_p^{tk-el-l(l-1)/2} C_p(t, k, j + 6m) Q^{-(x+1/2)e-(1+3x)l-wk-6zm}.$$

The four positive- t families are $(e, l) = (0, 2r + 2), (1, 2r), (0, 2r + 1), (1, 2r + 1)$. Increasing both r and m by one multiplies their summands by $\mathcal{R}$; further increases of m give V . Their boundary values give (A.3), using (A.2) and $\gamma_2 \gamma_4 = 1$. The $t = 0$ terms give the two displayed geometric terms outside P_p^* .

Finally, with the original zeros at $p \mid u$, put

$$W = \chi_p(u) Q^{-w}, \quad D = \eta(p) \overline{\chi_p(u)} Q^{-x}, \quad H_p = P_p \frac{(1-V)(1-W)}{1-D}, \quad \mathcal{H}_{\eta, u} = \prod_{p \notin S} H_p.$$

Summing these factors gives (6); the Poisson calculation then proves (5). Thus both representations refer to the same physical probe, including the nonsquarefree and shared-prime cases.

B Proof of the weak reflection formula

Proof of Lemma 2. Write $\mathbb{H}^3 = \{(z, v) : z \in \mathbb{C}, v > 0\}$, with its standard $\mathrm{SL}_2(\mathbb{C})$ action, and put $\check{e}(z) = \exp(2\pi i(z + \bar{z})) = e(\lambda z)$, $L(t) = \begin{pmatrix} 1 & 0 \\ t & 1 \end{pmatrix}$, and $E = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$. We use the cubic theta function of [2, (5.6)–(5.8)], normalized by

$$\theta(z, v) = \frac{3^{5/2}}{2} v^{2/3} + \sum_{\mu \neq 0} t_\theta(\mu) v K_{1/3}(4\pi|\mu|v) \check{e}(\mu z).$$

Here $K_{1/3}$ is the modified Bessel function of the second kind. Its coefficients are those of the cited formula (5.7); in particular, $t_\theta(\lambda^{-3}cb^3) = 3^{5/2}|b| \overline{\check{g}_3(c)}$ for primary squarefree c and primary b . For each cusp representative H , define the dual coefficients by

$$\overline{\theta(H(z, v))} = \overline{\sigma_H} v^{2/3} + \sum_{\mu \neq 0} d_H(\mu) v K_{1/3}(4\pi|\mu|v) \check{e}(\mu z). \quad (\text{B.1})$$

For $H = I, L(\lambda), L(-\lambda)$ these are respectively the conjugate expansions with indices $j = 1, 19, 10$ in [2, (5.9), (5.12), (5.16) and Appendix A]. More precisely, $d_H(\mu) = \overline{d_j^{\mathrm{DR}}(-\mu)}$. The coefficient $d(\mu)$ in each reflection branch is this $d_H(\mu)$.

Sextic reciprocity, including its zeros, writes the character of the completed index A as

$$\nu(A)\chi_A(m) = \Psi_0(A)\chi_R(A) \prod_{p|P, p \notin S} \chi_p(A)^{j_p}, \quad j_p \equiv v_p(P) \pmod{6}.$$

The fixed character Ψ_0 belongs to a finite family: the valuations at S matter only modulo six. Choose one sufficiently divisible fixed modulus L for that family, the primary condition and the cubic supplementary symbol. The function $\phi(x) = \chi_x(\lambda)^2 \Psi_0(x)$ on primary elements prime to S , zero elsewhere, is L -periodic. For $h_0 \in \mathcal{O}/L\mathcal{O}$, put

$$\widehat{\phi}(h_0) = q_L^{-1} \sum_{x \bmod L} \phi(x) e(-h_0 x/L), \quad \phi(x) = \sum_{h_0 \bmod L} \widehat{\phi}(h_0) e(h_0 x/L).$$

The pairing is nondegenerate and $|\widehat{\phi}(h_0)| \leq 1$. At the moving primes, finite Fourier inversion gives

$$C_{p,j}(h) = q_p^{-1} \sum_{x \bmod p} \chi_p(x)^j e(-hx/p) = \begin{cases} q_p^{-1/2} \tau_{p,j}^- \chi_p(h)^{-j}, & j \neq 0, h \neq 0, \\ 0, & j \neq 0, h = 0, \\ -q_p^{-1}, & j = 0, h \neq 0, \\ 1 - q_p^{-1}, & j = 0, h = 0, \end{cases}$$

where the nontrivial normalized Gauss factors have modulus one. Thus the required masked theta function is a finite sum of translates by

$$z_h = \lambda^2 \left(h_0/L + \sum_{p|DR} h_p/p \right), \quad h_p \neq 0.$$

All primes of R are active; only frozen primes with $j_p = 0$ can be inactive. This accounts for the number of terms.

This theta vector has precisely the coefficients in (7). Indeed, the primary condition on $x = \lambda^3 \mu$ isolates the infinity indices $\mu = \lambda^{-3} c b^3$. The other ramified exponents make x divisible by λ , and the negative branch gives $x \equiv -1 \pmod{3}$. The published conjugate-theta coefficient at the retained index is $3^{5/2} |b| \widetilde{g}_3(c)$, where $\widetilde{g}_3(c) = \chi_c(\lambda)^{-2} \gamma_2(c)$ on the retained support. The supplementary factor in ϕ cancels this symbol. Its whole-index zero mask removes exactly the indices meeting S , and every moving character still acts on $c b^3$ with its zeros.

The published theta function satisfies $\theta(gw) = \kappa(g)\theta(w)$ on $\Gamma_2 = \langle \mathrm{SL}_2(\mathbb{Z}), \Gamma_1(3) \rangle$, where $\Gamma_1(3) = \{g \in \mathrm{SL}_2(\mathcal{O}) : g \equiv I \pmod{3}\}$ and $\kappa(g) = (c/a)_3$ on $\Gamma_1(3)$ for $c \neq 0$; it is one when $c = 0$ and on $\mathrm{SL}_2(\mathbb{Z})$.

Here are the rational-cusp details needed to make the dual coefficients common to the rows. Reduce $\lambda^2 h_0/L = a_F/c_F$. Multiply this fixed pair by a unit so that $a_F \equiv 1 \pmod{3}$ if $\lambda \mid c_F$, and $c_F \equiv 1 \pmod{3}$ otherwise. This is possible because reduction identifies μ_6 with $(\mathcal{O}/3\mathcal{O})^\times$; the selected entry is a unit modulo 3. The choice depends only on h_0 . Put $r = DR$, and write

$$z_h = a/c, \quad c = c_F r, \quad a = a_F r + \lambda^2 c_F \sum_{p|r} (r/p) h_p.$$

This fraction is reduced. Take $M = \lambda^{12} L^4$, lift h_p to be zero modulo M^2 , and fix $r \bmod M^2$. Complete a/c to $g = \begin{pmatrix} a & b_0 \\ c & \delta \end{pmatrix}$: choose $\delta \equiv a^{-1}$ at the prime-power parts of $M c_F$ supported on c_F , $\delta \equiv 0$ at the other prime-power parts of M , and $\delta \equiv a^{-1} \pmod{p}$ for $p \mid r$. Then $b_0 = (a\delta - 1)/c$, and all fixed-prime matrix residues are independent of the active h_p . Since r is primary and the

moving terms contain λ^2 , a is primary when $\lambda \mid c$, and c is primary otherwise. The CRT choices then give $g \equiv I \pmod{3}$ if $\lambda^2 \mid c$, $g \equiv L(u_0) \pmod{3}$ with $u_0 = \pm\lambda \equiv c \pmod{3}$ if $v_\lambda(c) = 1$, and $g \equiv \begin{pmatrix} u_0 & -1 \\ 1 & 0 \end{pmatrix} \pmod{3}$ with $u_0 \equiv a \pmod{3}$ otherwise. Use these as H ; hence $gH^{-1} \in \Gamma_1(3)$. In the last case $H = EL(-u_0)$ and its theta function is one of the same three, since $E \in \text{SL}_2(\mathbb{Z})$ and lower translations by $\mathbb{Z} + 3\mathcal{O}$ have multiplier one. The fixed unit changes no rational translate or norm scale and only multiplies the scalar by unit phases.

The published automorphy multiplier and cubic reciprocity give

$$\kappa(gH^{-1}) = \kappa_F \prod_{p \mid r} \chi_p(a)^2, \quad (\text{B.2})$$

The three theta expansions are those of $I, L(\lambda), L(-\lambda)$, since $\lambda - \omega^2, -\lambda - \omega \in \mathbb{Z} + 3\mathcal{O}$. For explicit verification, the fixed factors in the cases $v_\lambda(c) \geq 2$, $v_\lambda(c) = 1$, and $v_\lambda(c) = 0$ are respectively

$$(c_F/a)_3, \quad (-u_0/(a - u_0 b_0))_3 ((c_F/u_0)/a)_3, \quad (a/c_F)_3.$$

In the middle case $u_0 = \pm\lambda$ is congruent to c modulo 3; use $a(c - u_0\delta) = c(a - u_0 b_0) - u_0$ and $b_0 c \equiv -1 \pmod{a}$. In the last case the CRT choice gives $\delta \equiv 0 \pmod{9}$, hence $-b_0 c = 1 - a\delta \equiv 1 \pmod{9}$. Factor δ into its unit, λ -power, and primary part. Against the denominator $1 - a\delta$, the supplementary laws make the first two factors trivial; reciprocity makes the primary part trivial because $1 - a\delta \equiv 1 \pmod{\delta}$. Thus $(\delta/(-b_0 c))_3 = 1$, and

$$\kappa(gH^{-1}) = (\delta/(-b_0))_3 = (\delta/c)_3^{-1} = (a/c)_3 = (a/c_F)_3 (a/r)_3.$$

The chosen residues make each fixed factor independent of the active frequencies. Periodic changes of the cusp representative and of δ avoid zero matrix entries in these symbol formulas without changing any congruence.

Write $k_g = gH^{-1} \in \Gamma_1(3)$. Since $g = k_g H$, $\overline{\theta(w)} = \overline{\kappa(k_g)} \overline{\theta(Hg^{-1}w)}$. For $w = (a/c + z, v)$ the coordinates of $g^{-1}w$ are

$$z' = -\frac{\delta}{c} - \frac{\bar{z}}{c^2(|z|^2 + v^2)}, \quad v' = \frac{v}{q_c(|z|^2 + v^2)}.$$

At $z = 0$, differentiation in $\bar{z}$ gives $\partial_{\bar{z}} z' = -(cv)^{-2}$ and $\partial_{\bar{z}} \bar{z}' = \partial_{\bar{z}} v' = 0$. It therefore becomes the z' derivative of the dual cusp series, at height $1/(q_c v)$, and kills every cusp constant.

Put $x = \lambda^4 \mu$ and $f = \lambda^3 c_F$. Let δ_F be the residue of δ modulo f , and let r^{-1} denote the inverse of r modulo f . For $p \mid r$, put $\epsilon_p \equiv -(\lambda^5 c_F^2 (r/p)^2)^{-1} \pmod{p}$. Since $a \equiv \lambda^2 c_F (r/p) h_p \pmod{p}$ and $\delta \equiv a^{-1} \pmod{p}$, additive CRT gives

$$\check{e}(-\delta\mu/c) = \vartheta(x) \prod_{p \mid r} e(\epsilon_p h_p^{-1} x/p), \quad \vartheta(x) = e\left(-\frac{\delta_F r^{-1} x}{f}\right).$$

The residue δ_F and r^{-1} modulo f are fixed in the chosen sector. Thus this is a common column phase. Conjugating (B.2) contributes $\chi_p(h_p)^{-2}$. Combining with $C_{p,j}$ and putting $y = h_p^{-1}$ gives a Gauss sum of exponent $j + 2$. At a prime of R , $j = 1$, so its column factor is exactly $\chi_p(x)^{-3} = \chi_p(x)^3$. At a frozen active prime it is

$$B_p(x) = \begin{cases} \chi_p(x)^{-j_p-2}, & j_p \notin \{0, 4\}, \\ q_p^{-1/2}(-1 + q_p 1_{p \mid x}), & j_p = 4, \\ q_p^{-1/2} \chi_p(x)^{-2}, & j_p = 0. \end{cases}$$

The factors involving the units ϵ_p contribute only scalars of modulus one, absorbed into ζ_R ; in a Ramanujan sum they leave the condition $p \mid x$ unchanged. Thus the displayed B_p are independent of R , include all collisions, and satisfy the asserted bound.

For the normalization, let $D_m(v)$ denote the $\bar{z}$ derivative at $z = 0$ of the masked conjugate theta series constructed above, and let $\mathcal{D}_m(t)$ denote the series in (7) with $V(q_c q_b^3/Z)$ replaced by $(q_c q_b^3)^{-t}$. At $\mu = \lambda^{-3} c b^3$, $\bar{\mu} = -i |c b^3| \overline{\alpha(c b^3)} / (3\sqrt{3})$. The differentiated coefficient is consequently $6\pi |c b^3| |b| \gamma_2(c) \overline{\alpha(c b^3)}$ times the original character masks. Using $\int_0^\infty K_{1/3}(y) y^{w-1} dy = 2^{w-2} \Gamma((w - 1/3)/2) \Gamma((w + 1/3)/2)$ for $\text{Re } w > 1/3$ gives

$$\int_0^\infty D_m(v) v^{2t} dv = \frac{81}{8\pi} \left(\frac{27}{4\pi^2} \right)^t \prod_{\pm} \Gamma(1 + t \pm 1/6) \mathcal{D}_m(t) \quad (\text{B.3})$$

first for $\text{Re } t$ sufficiently large. In the reflected integral write $c_{\text{cusp}} = c_F D R$ to distinguish the cusp denominator from the squarefree summation variable c . The substitution $v' = 1/(q_{c_{\text{cusp}}} v)$ contributes $q_{c_{\text{cusp}}}^{1-2t}$, and the differentiated automorphy factor contributes $-c_{\text{cusp}}^{-2}$. The absolute conductor factor cancels because $c_{\text{cusp}}^{-2} q_{c_{\text{cusp}}} = \overline{\alpha(c_{\text{cusp}})}^2$. The dual Bessel integral has scalar $i/(8\pi)$. Dividing by (B.3) and replacing t by $-t$ therefore gives the kernel

$$V^\sharp(x) = \frac{1}{2\pi i} \int_{(\sigma)} e^{t^2} \prod_{\pm} \frac{\Gamma(1 + t \pm 1/6)}{\Gamma(1 - t \pm 1/6)} \left(\frac{(2\pi)^4 x}{27} \right)^{-t} dt, \quad \sigma \geq 0.$$

Before local summation the scalar is $-i \overline{\alpha(c_{\text{cusp}})}^2 / 81$; the remaining conductor power is entirely inside $V^\sharp(q_\mu Z / q_{c_{\text{cusp}}}^2)$. Every remaining scalar is a residue-symbol unit, a normalized Gauss sum, $\widehat{\phi}(h_0)$ of modulus at most one, or an inactive factor $1 - q_p^{-1}$. This proves $|\zeta_R| \leq 1/81$. The three published cusp formulas give the stated support and coefficient bound.

The differentiated translates decay exponentially at both ends of their height integral, by the direct and rational-cusp Fourier expansions. Their Mellin transform is entire and bounded on every closed vertical strip. Dividing by the gamma factors in (B.3), Stirling's formula bounds $\mathcal{D}_m(t)$ on each such strip by $C_m(1 + |\Im t|)^A e^{\pi |\Im t|}$. Multiplication by e^{t^2} then makes the horizontal integrals tend to zero and justifies the Mellin shift used above. Stirling on any fixed nonnegative line, with the Gaussian, bounds every Euler derivative of $V^\sharp$; moving that line arbitrarily far right proves its rapid decay. All dual sums are absolutely convergent. $\square$

Use of AI: The author's main contributions are the following insights:

- Replace collision estimates with cubic theta reflection and quadratic Hecke mean-square bound.
- Replace the detector construction with fixed-height zero bins and Hecke zero-density theorem.
- Identify the principal row using CRT and conductor analysis. Avoid Chebotarev.
- Extract the principal signal through two successive residues. Control the remaining terms with uniform contour and tail estimates.

The author used Claude and ChatGPT to summarize and understand OpenAI's proof and then prompted his insights to work out the details interactively and simply the proofs one by one. Proofs generated by Claude were verified by Codex and vice-versa. The author verified the outputs from both and combined them into one unified proof using ChatGPT.

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